

Student Answer Script View



MIT Bangalore MID-TERM 2025-2026

Answer Sheet

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Comments submitted by student.

Q.No : 1)

Score : 1.00 / 1.00

Image processing hardware in the system is mainly used to

Provide storage for large image datasets

Perform parallel arithmetic and logical operations on images

Convert analog images into digital images

Display processed images

Q.No : 2)	Score : 1.00 / 1.00
If an image is transformed using the negative transformation $s=L-1-r$ which of the following applications benefits most?	
Enhancing details in dark regions of medical X-rays	Improving efficiency of JPEG compression
Equalizing histogram of natural photographs	Compressing dynamic range of astronomical images

Q.No : 3)

Score : 0.00 / 1.00

An image of size 1024×1024 pixels is quantized to 16 gray levels. How many bits are required to store the image?

☐ $1024 \times 1024 \times 4$ ☐ $1024 \times 1024 \times 8$ ☒ $1024 \times 1024 \times 16$ ☐ $1024 \times 1024 \times 2$

Q.No : 4)		Score : 1.00 / 1.00
Compression in image processing is mainly useful for		
Enhancing contrast	Storing and transmitting images efficiently	Reducing segmentation errors
Detecting shapes in images		

Q.No : 5)Score : 0.00 / 1.00

Which statement best explains the practical difference between RGB and HSV models?

HSV is device-dependent, while RGB is perceptually uniform.	RGB is always superior for histogram equalization tasks	HSV is better for segmentation because hue is invariant to illumination changes, unlike RGB	HSV is device-dependent, while RGB is perceptually uniform.
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Q.No : 6)

Score : 1.00 / 1.00

In digital image processing, a pixel represents:

Image storage unit

Smallest element of an image

Image segmentation method

Wavelet coefficient

Q.No : 7)

Score : 1.00 / 1.00

Specialized modules and user-written code for processing images belong to:

Image Processing Software

Image Display Device

Mass Storage

Hard Copy Device

Q.No : 8)

Score : 1.00 / 1.00

Which of the following transformations is most appropriate to enhance details in dark regions of a medical image?

Logarithmic transformation.

Negative transformation

Power-law with $\gamma > 1$

Bit-plane slicing of the MSB

Q.No : 9)

Score : 1.00 / 1.00

In industrial applications, thermal imaging is primarily used for:

Crop monitoring Detecting material defects Identifying emotions Enhancing entertainment videos

Q.No : 10)

Score : 0.00 / 1.00

Which component provides a general-purpose platform for running image processing tasks?

☒ Image Sensor☐ Computer☐ Hard Copy Device☐ Network

Q.No : 11 A)

Score : 1.00 / 5.00

Analyze the role of Power-Law Transformation and Logarithmic Transformation in gray-level image enhancement. With the help of suitable graphs, discuss how each transformation modifies the dynamic range of dark and bright regions, and explain why such enhancements are necessary in industrial quality inspection of materials and defect detection.

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Logarithmic Transformation: $S = c \cdot \log(1+r)$
Power - Law transformation: $S = c r^{\gamma}$

Answer: The role of power-law transformation and Logarithmic transformation is to enhance gray images.

Q.No : 11 B)

Score : 3.00 / 3.00

A hospital uses computer vision to automatically detect tumors in medical scans. Describe how each step of digital image processing is applied in this system.

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Answer:- In digital image processing A hospital uses the applications of computer vision to detect tumors in medical scans it includes steps like

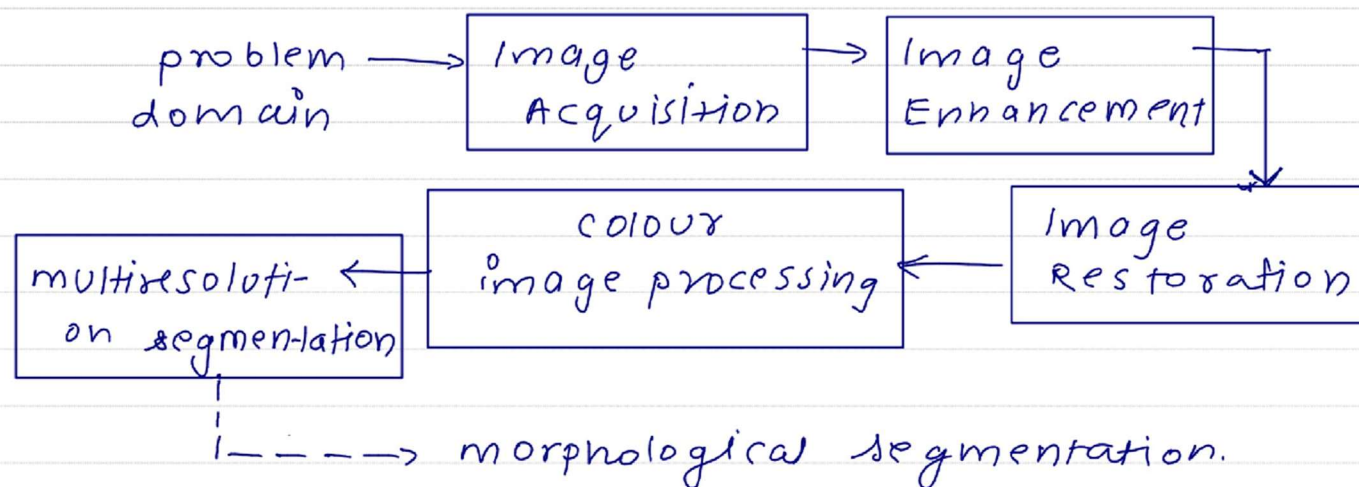


Image Acquisition: First step to detecting tumors in medical scans is image acquisition. The image is first fetched in the computer system before it can be processed, and converted from analog image to digital image.

Image Enhancement: Next step is that the image once converted into digital form is then enhanced with appropriate Transformers, for our case to detect tumors in medical scans it is passed through image negative filter to darken the tumor to make it more visible.

Image Restoration: The filtered image of the scan is then processed for

fixing the blurred edges, for our case, we use spatial filters to do fix smoothing to restore the image for a perfect fit for our application of computer vision to detect tumor

colour image processing : In this step the RGB is calibrated for darkening out the of any possible signs of tumor

multiresolution segmentation : In this step images are segmented for a visible lining of the tumor inside the scanned image.

Q.No : 11 C)

Score : 2.00 / 2.00

Apply image negation on the following image of size 4x4 and write the output image matrix.

$$I = \begin{bmatrix} 0 & 50 & 128 & 255 \\ 30 & 90 & 180 & 220 \\ 10 & 40 & 60 & 200 \\ 15 & 100 & 150 & 240 \end{bmatrix}$$

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Solution:- $I =$

0	50	128	255
30	90	180	220
10	40	60	200
15	100	150	240

$$\text{Range} = [0 - 255]$$

For Image Negation Truth Table
For NOT gate. $A = \neg A$

A	$\neg A$
0	1
1	0

Levels

$$2^8 = 256$$

 L^k $\therefore 256 \text{ Levels}$

$$S = (L-1) - r$$

$$= (256-1) - r$$

$$S = 255 - r$$

$$S = \begin{cases} 255 - r & r < 0 \\ & r > 1 \end{cases}$$

$$I' =$$

225	205	127	0
225	165	75	35
245	215	195	55
240	155	105	15

$$[255 - r]$$

Range

output matrix

Q.No : 12 A)

Score : 5.00 / 5.00

Consider the following image matrix. Compute the smallest 4-, 8 and m-path in the given image matrix and also mention the path length. Consider $V = \{2, 3, 5\}$.

4	5	6	4	5	4
3	5	3	4	3	4
4	4	4	2	4	3
5	3	4	3	3	5
6	3	2	3	4	6
5	4	5	3	3	6

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Answer :

4	5	6	4	5	4
3	5	3	4	3	4
4	4	4	2	4	3
5	3	4	3	3	5
6	3	2	3	4	6
5	4	5	3	3	6

$$V = \{2, 3, 5\}$$

The smallest path that exist between (p and q) for m-adjacency is

				3	
			2		
			3		
3	2	3			

3	2	3	3	2	3
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6 units for m-adjacency

Let us compute for D_g (for 4 Adjacency)

Answer :

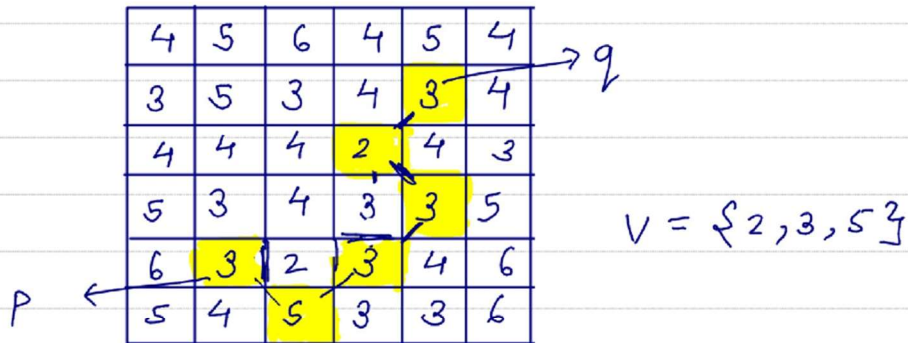
4	5	6	4	5	4
3	5	3	4	3	4
4	4	4	2	4	3
5	3	4	3	3	5
6	3	2	3	4	6
5	4	5	3	3	6

$$V = \{2, 3, 5\}$$

There exist no path for 4 Adjacency as

the co-ordinate q is surrounded by the number 4 that does not exist in $\gamma = \{2, 3, 5\}$

Now Let us check,
For D₈ adjacency



for 8 adjacency the smallest path possible
is 6 units.

3	5	3	3	2	3
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Answer key : 4- Adjacency path: ∞ units
8- Adjacency path: 6 units
m- Adjacency path: 6 units

Q.No : 12 B A)

Score : 0.50 / 3.00

An automobile company uses an automated vision system to inspect defects in car parts. The system captures part images, processes them to detect scratches or cracks, stores inspection results, and generates quality reports. Identify and explain the role of each component in the image processing system in this scenario.

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Solution:- Automobile vision system

- i) captures Images: Image Acquisition detection
- ii) detects cracks : Image processing & object
- iii) Generates : computer intelligence
quality reports

i) captures Images: When images are captured image acquisition is taken in role. The images of the car parts are converted into analog to digital.

ii) detects cracks: cracks in the parts are detected using image processing to make the cracks visible and, cracks are treated like objects to detect cracks in the automobile parts

iii) Generates quality reports : The quality reports are generated using computer intelligence, which is based on a machine learning model that classifies cracks of parts and healthy parts in an automobile industry.

Q.No : 12 B B)

Score : 0.50 / 2.00

Discuss how pixel intensity values are mapped to discrete levels and its effect on image quality.

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Answer: pixel intensity values are mapped to discrete level affect its image quality because:-

i) Intensity is defined as $I = L^K$ where
I: Intensity K: no. of bits per dpi
L: Levels

ii) As we increase the intensity we are increasing the $\log_K(L)$ value of the image, The logarithmic value of L increases which increases intensity thus affecting the image quality

iii) pixel intensity values are mapped to discrete Levels by $I = L^K$ where
L: Levels K: no. of bits I is pixel intensity.



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